

Creditor recovery: The macroeconomic dependence of industry equilibrium☆

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Highlights

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State of economy drives asset liquidation values and therefore recovery rates.
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Macroeconomic effects induce industry downturns and possible distress.
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Industries whose sales are more correlated with GDP recover less during recessions.
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Industries more dependent on external finance recover less when stock market falls.
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Provides a framework for macroeconomic stress testing of corporate credit.

Abstract

This paper reconciles the state of the economy with industry conditions in driving asset liquidation values and, therefore, recovery rates on defaulted debt securities. Evidence to date downplays the economywide effect in favor of industry and debt characteristic explanations. This paper shows that macroeconomic effects are important but operate differentially at the industry level. Industries whose sales growth is more correlated with GDP growth recover less during recessions. And industries that are more dependent on external finance recover less when the stock market falls. These findings expose how economywide shocks are transmitted to industry

downturns, providing a framework for the role of aggregate risk in recovery risk and for macroeconomic stress testing.

Introduction

In the aftermath of the recent financial crisis, both practitioners and academics have paid increasing attention to creditor recovery rates (or its opposite, loss given default). This paper investigates whether creditor recoveries depend on the macroeconomy in an economically important manner. This research reconciles two different strands of the literature on corporate recovery rates. One strand explicitly models the dependence of recovery rates on the business cycle. A primary example is the Federal Reserve's annual stress testing of large financial institutions, whereby the likelihood of a large loss is explicitly conditioned on adverse macroeconomic outcomes.¹ The other strand, mainly academic-focused research, increasingly downplays economywide effects. These studies—discussed at greater length in what follows—find that business-cycle conditions do not matter once one accounts for specific industry, bond market, and control-rights explanations.

This paper's contribution generally is to bridge these two strands of the literature by tying industry downturns with the state of the economy. The academic literature does not allow for the possibility that the extent of industries' sensitivity to the business cycle varies, whether for fundamental or liquidity reasons. This paper questions whether industry shocks are exogenous, as assumed, and considers instead whether economywide shocks induce industry downturns and possible industry fire sales. The paper's main contribution is in showing that different industries have different exposures to the business cycle and the stock market, and therefore, industry controls can mask such aggregate effects. This study, therefore, provides a framework for macro-based stress testing of recovery risk.

A stylized fact that motivates the first strand of the literature is the observation that the recovery rate is inversely related to the aggregate default rate (Fig. 1). The time series shown in Fig. 1 are aggregate default rates and recovery rates, weighted by the debt amount, following the methodology in Altman and Kuehne (2011).² Defaults were clustered during 1982, the early 1990s, the early 2000s, and 2008–09—periods of low recovery rates. Indeed,

Giesecke et al. (2011) highlights the clustered nature of corporate bond default events at various times over the longer historical period they examine, including the railroad crisis of 1873–75, the banking panics of the late 1800s, and the Great Depression.

One reason for the inverse relation between the recovery rate and the default rate is common dependence on a systematic factor such as the business cycle. For example, the same adverse economic conditions that cause defaults to rise also depress recoveries. Evidence that the recovery rate is pro-cyclical is shown in Fig. 2, where the shaded areas are recessions. For example, the recovery rate is positively correlated with real GDP growth (correlation coefficient equal to 0.32 over 1978–2010). Previous studies in this first strand of the literature also document a similar macroeconomic dependence, whereby recessions depress bond recoveries by up to one-third from normal-year averages (Frye, 2000, Schuermann, 2005; Fig. 2).

Recent academic literature, however, favors industry and other debt characteristic explanations. A common thread in this strand of the literature is the observation that there is only, at best, “a loose coupling of loss severity to the business cycle,” as noted by Schuermann (2014) in his review of the literature. There are several competing alternative explanations, including Acharya et al. (2007). In that paper, conditions in the industry of the defaulting firm are important for recovery on the firm's real assets. In an alternative paper by Altman et al. (2005), conditions in the distressed bond market are important for recovery on the firm's securities because the capacity of financial investors to absorb defaulted securities is limited.

In these and other papers, industrywide downturns contribute to a lower recovery rate through two main channels: fundamentals and illiquidity. First, times of reduced business opportunities in a firm's industry are invariably reflected in lower asset liquidation values. Intuitively, creditor recoveries will depend on the value of the debt collateral. But the collateral, and more generally the economic worth of the defaulted firm's assets, are revised downward in line with an industry downturn. Second, an industry downturn can impose an additional fire-sale discount on the defaulted firm's assets, beyond its effect on fundamental value. In this view, would-be buyers of liquidated assets are likely to be the defaulted firm's industry peers who also

are likely financially distressed and unable to buy the liquidated assets. Therefore, distress in the firm's industry limits its peers from bidding on the defaulted firm's assets up to their "value in best use" when managed by industry specialists. Shleifer and Vishny (1992) first developed this market equilibrium approach and described the difference between the price and the best-use value of an asset as "asset illiquidity."³ Both effects (industry fundamentals and distress) limit the amount that creditors are able to recover from realized or *anticipated* liquidations of a firm's assets.

While these papers focus on industry-specific shocks, a logical consideration is that industry shocks may be endogenous and triggered by an overall weak economy. Indeed, in the original model by Shleifer and Vishny (1992), the shock is allowed to be economywide, not simply industrywide, so that a recession hits some firms harder. The important point is that shocks are not specific to the asset seller (i.e., not idiosyncratic in nature).⁴ This paper's main contribution is therefore to allow for the transmission from the macroeconomy through either or both the economic-worth channel and the fire-sale channel. An economywide recession reduces business opportunities, lowering the economic worth of defaulted firms' assets. For example, business prospects may be more sensitive to economic downturns in some industries, such as apparel and accessories, than in others, such as tobacco. Similarly, a recession can increase susceptibility to industry distress, lowering the liquidation value of assets in some industries further below their revised economic worth.

The results of this paper show that macroeconomic effects do operate differentially via industry conditions, supporting the main hypothesis, which is that the effect of economywide shocks is not necessarily uniform. The analysis draws on about 40 years of recovery data on defaulted debt instruments and controls for a range of contract, firm, and industry characteristics. Specifically, the paper finds that industries with a greater correlation of their sales growth and GDP growth have a smaller recovery during recessions. For example, in a recession, the industry at the 75th percentile recovers roughly seven cents less on the dollar than the industry at the 25th percentile. This is an economically significant effect as it is comparable to the effect of a proxy for industry distress of seven cents to 10 cents (and I confirm that the industry distress effect closely matches that in

Acharya et al., 2007). Similarly, recoveries in industries highly dependent on external finance increase with positive movements in the stock market. Moreover, the effects of direct proxies for industry distress and economic worth become weaker or insignificant once different industries are allowed to respond differently to the macroeconomy. Note that it is reasonable in many cases that industry conditions may be independent of the state of the economy, such as episodes of accounting fraud or regulatory and technological changes (Covitz and Han, 2004). The point simply is that industry conditions triggered by the economy are also important (yet, so far overlooked) drivers.⁵ Finally, the macro-sensitive industry channel is robust to the inclusion of bond market conditions as well as macroeconomic conditions at the time of debt origination, as respectively emphasized in previous studies (e.g., Altman et al., 2005, Zhang, 2010). I also show that, in addition to the commonly examined factors of GDP growth and stock market returns, other macroeconomic factors are associated with hypothesized movements in industry conditions. These include house prices, interest rates, and credit spreads.

It is interesting to consider the parallels of this research with a different research topic that examines the determinants of cyclical fluctuations in the unemployment rate—whether these are due to aggregate demand shocks or to industry shifts. Abraham and Katz (1986) observe that aggregate demand fluctuations can be primarily responsible, even if the unemployment rate is positively related to cross-industry dispersion of employment growth. The idea also is that sectors differ in their sensitivity to the common aggregate demand shock.

The rest of this paper is organized as follows: Section 2 reviews descriptive statistics of creditor recoveries together with the related literature; Section 3 describes the empirical methodology employed; Section 4 examines the macroeconomic determinants of recovery rates, considering the role of industry distress and their interrelationships; Section 5 follows with additional results and robustness checks; and Section 6 concludes with a focus on financial intermediary loan losses and policy implications for financial stability.

Section snippets

The determinants of recovery rates and related literature

Analysis of credit risk has been devoted mostly to the likelihood of default with less attention paid to recovery or loss given default (e.g., the review in Altman et al., 2004). A common assumption is to fix recovery based on historical averages, such as between 40% and 50% on debt issued by U.S. corporate borrowers and 25% on debt issued by sovereign borrowers (Das and Hanouna, 2009). For example, Giesecke et al. (2011) apply a long-run average loss rate of 50% and devote their analysis to

The empirical framework and data

The basic multivariate empirical specification relates recovery rates on individual debt instruments to the macroeconomic state and the dependence of the defaulted borrower's industry on the macroeconomy: where *Recovery* is the recovery rate on debt instrument i issued by borrower j . The coefficient vector θ captures the economic effect of the borrower's industry sensitivity to the

Evaluating industry conditions and the macroeconomy separately

The direct effect of industry distress and industry Q on recovery rates is shown in the first two columns of Table 3. Widespread financial distress in a particular industry is found to depress the recovery rate by 10.62 percentage points in column (1), supporting an industry illiquidity view, controlling for industry Q and industry dummies in addition to other contract- and firm-level variables. The effect attributed to time-varying industry distress is even greater when not controlling for

Additional results, robustness tests, and model performance

Other macroeconomic factors such as house prices, interest rates, and credit spreads are associated with hypothesized movements in industry conditions. The results are presented in Table 7 (in which all the contract- and firm-level

controls are also included in the regressions but not shown in the interest of space). Care should be taken not to overinterpret these results, especially as examination of these factors is less common in this literature than are growth and stock returns.

Conclusion and financial stability implications

This paper pinpoints how industries that are most susceptible to a macroeconomic downturn yield depressed creditor recoveries. While intuitively appealing, empirical evidence to date has not considered this economywide transmission to (anticipated) asset liquidation values in different industries.

The significant dependence of asset liquidation values and recovery rates on the business cycle has important implications for risk management at banks and other financial intermediaries. The purpose

References (42)

- V.V. Acharya *et al.*
[Does industry-wide distress affect defaulted firms? Evidence from credit recoveries](#)
J. Financ. Econ.
(2007)
- R.L. Bennett *et al.*
[The effects of resolution methods and industry distress on the loss on assets from bank failures](#)
J. Financ. Stab.
(2014)
- M. Bruche *et al.*
[Recovery rates, default probabilities, and the credit cycle](#)
J. Bank. Finance
(2010)
- J.D. Coval *et al.*
[Asset fire sales \(and purchases\) in equity markets](#)
J. Financ. Econ.
(2007)
- S.R. Das *et al.*

[Implied recovery](#)

J. Econ. Dyn. Control
(2009)

- K. Giesecke *et al.*

[Corporate bond default risk: a 150-year perspective](#)

J. Financ. Econ.
(2011)

- R. Jankowitsch *et al.*

[The determinants of recovery rates in the US corporate bond market](#)

J. Financ. Econ.
(2014)

- E. Jokivuolle *et al.*

[Cyclical default and recovery in stress testing loan losses](#)

J. Financ. Stab.
(2013)

- H.D. Khieu *et al.*

[The determinants of bank loan recovery rates](#)

J. Bank. Finance
(2012)

- T. Schuermann

[Stress testing banks](#)

Int. J. Forecast.
(2014)

[View more references](#)

Cited by (19)

- [Interpretable machine learning for creditor recovery rates](#)

2024, Journal of Banking and Finance

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- [Intertemporal defaulted bond recoveries prediction via machine learning](#)

2022, European Journal of Operational Research

Citation Excerpt :

Evidence for the importance of macroeconomic variables in credit risk management can be found in Bruche & Gonzalez-Aguado (2010); Chava, Stefanescu, & Turnbull (2011); Jankowitsch et al. (2014); Mora (2015); Nazemi et al. (2017); Varma & Cantor (2005), and Nazemi & Fabozzi (2018). We use the database from the Federal Reserve Bank of St. Louis (FRED, Federal Reserve Economic Data) complemented by aggregate default data from Fitch Ratings to retrieve 182 macroeconomic variables used in the credit risk literature such as Acharya et al. (2007); Jankowitsch et al. (2014); Mora (2015); Varma & Cantor (2005), and Nazemi & Fabozzi (2018). The macroeconomic data covers the time period from 2000 (one year before the start of the recovery rate observation period) until 2016.

Show abstract

- [Predicting loss given default using post-default information](#)
2021, Knowledge Based Systems

Citation Excerpt :

For corporate bonds, the characteristics of collateral, firm capital structure and business cycles are all considered to be significant drivers according to [3]. Acharya et al. [14] find that industry distress indicators remarkably influence corporate bond LGD, and similar evidence can be found in [15]. For bank loans, the impact from loan level features has been shown to be more important.

Show abstract

- [Macroeconomic variable selection for creditor recovery rates](#)
2018, Journal of Banking and Finance

Citation Excerpt :

Similarly, this study looks to find a relationship between recovery rates and the four macroeconomic variables affecting loans. Mora (2015) incorporated several macroeconomic variables in her model: GDP growth, stock market return, housing price growth, housing price growth (state-level), 3-month Treasury bill rate, commercial paper spread (3-month commercial paper rate for high grade nonfinancial borrowers minus the 3-month Treasury bill rate) and corporate bond Baa-Aaa yield spread in recovery rate estimation for U.S. corporate bonds. She reports that macroeconomic variables are statistically significant in modeling recovery rates but they do not have the same effect on every industry.

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- [Fuzzy decision fusion approach for loss-given-default modeling](#)

2017, European Journal of Operational Research

Citation Excerpt :

Significantly lower recovery rates for defaulted firms if the industry of defaulted firms is in distress were reported by Acharya, Bharath, and Srinivasan (2007). However, Mora (2015) shows that (1) macroeconomic variables are important factors in LGD for corporate bonds, (2) industries which are more dependent on the global and national economies have lower recovery rates when the stock market drops, and industries whose sales are more related to GDP growth recover less during macroeconomic downturns. Both industry works and academic papers (such as Loterman et al., 2012; Qi & Zhao, 2011) sometimes use the non-parametric regression in LGD modeling.

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- [The social value of financial expertise](#)
2019, American Economic Review